Priority parking. Promised to members.

Members book access to a dedicated zone beside the B1 lifts. Licence-plate cameras log every entry and exit, so ParkFlow only accepts a booking when space is genuinely available.

See how the promise works

Review the business case

Proposed pilot site: KL East Mall. Forecasts guide walk-in access and arrival planning. Verified gate counts protect member bookings.



Sime Darby Property University Hackathon 2026 · Universiti Malaya

Team HEXA

Five Universiti Malaya students in Data Science, Actuarial Science, and AI—building practical car-park operations for Sime Darby Property.

B1-01

SE

HEXA 01

Leow Shen En

Team lead, product & UI/UX

Data Science

B1-02

PC

HEXA 02

Pohyi Chong

Forecasting & ML models

Data Science

B1-03

IT

HEXA 03

Isaac Teo

Data analysis & simulation

Data Science

B1-04

JS

HEXA 04

Lee Jia Shern

Business case & finance

Actuarial Science

B1-05

ZL

HEXA 05

Choong Zhuo Lin

AI engineering & prototype

Artificial Intelligence

Parking Problem

Saturday, 6:30 pm at KL East Mall. Your movie starts at 7:00, and you are still circling B1 hoping a bay opens up.



25 min

spent per day by the average Kuala Lumpur driver searching for a bay.

Have skipped a destination because parking was difficult — 40%

Have abandoned a trip and driven home — 26%

Want the option to book a bay ahead of arrival — 67%

For KL East Mall: Turn reliable weekend parking into a reason for loyal shoppers to return.

Sources: BCG, Unlocking Cities (2018); INRIX (2017); ParkMobile (2024). Industry benchmarks illustrate urban driver behaviour ahead of the KL East baseline survey.

Current Parking Options

Normal parking

- Uses every bay
- Zero certainty at peak

First come, first served

Drivers take available bays in arrival order.

Level counters only show the present

Signs report current availability, without securing future access.

Loyal members face the same queue

Members still face uncertainty during busy periods.

Member parking benefits

- Valued by members
- Different from timed booking

1 Utama ONECARD

Member parking zone with plate-recognition access.

Sunway Pyramid

Preferred parking closer to entrances at a higher rate.

The Gardens Club

Parking and valet benefits linked to membership tier.

Published programmes: 1 Utama ONECARD FAQ · Sunway Pyramid Parking FAQ · The Gardens Club Privileges.

How It Works

Book zone access, enter by verified plate, then park in any free bay.



Bookings reserve shared zone capacity. Individual bays are never assigned or locked.

Member Benefits & Pricing

BOOK

Eligible members

Reserve Priority Zone access before leaving home

PLAN

All members

Check peak-hour forecasts and expected turnover windows

ARRIVE

Registered plate

Drive straight in and out with automatic plate verification

GUIDE

General visitors

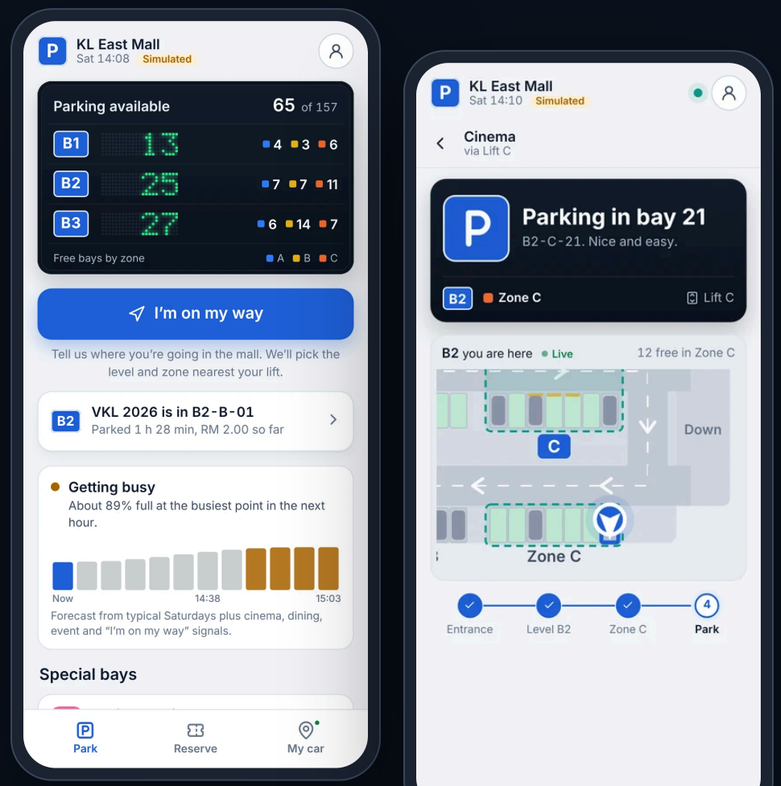
See live availability by floor while normal parking stays unchanged

RM4 per weekday booking

RM5 per peak weekend / public-holiday booking

Proposed booking fees, in addition to normal parking charges.

Eligible members only. Members without bookings may enter in Normal mode when spare capacity permits. Pilot rules are agreed with SDP.



Member App

1. **Check live parking conditions**
   See open bays by level and 30-to-90-minute busyness forecasts before setting off.
2. **Reserve Priority Zone access**
   When verified capacity is open, lock in a 30-minute arrival window in one tap.
3. **Drive in by registered plate**
   The zone camera verifies your number plate automatically—no paper tickets or QR scans at the ramp.
4. **Exit and release the slot**
   Driving out logs your departure and immediately frees capacity for the next member.

Open the member screen

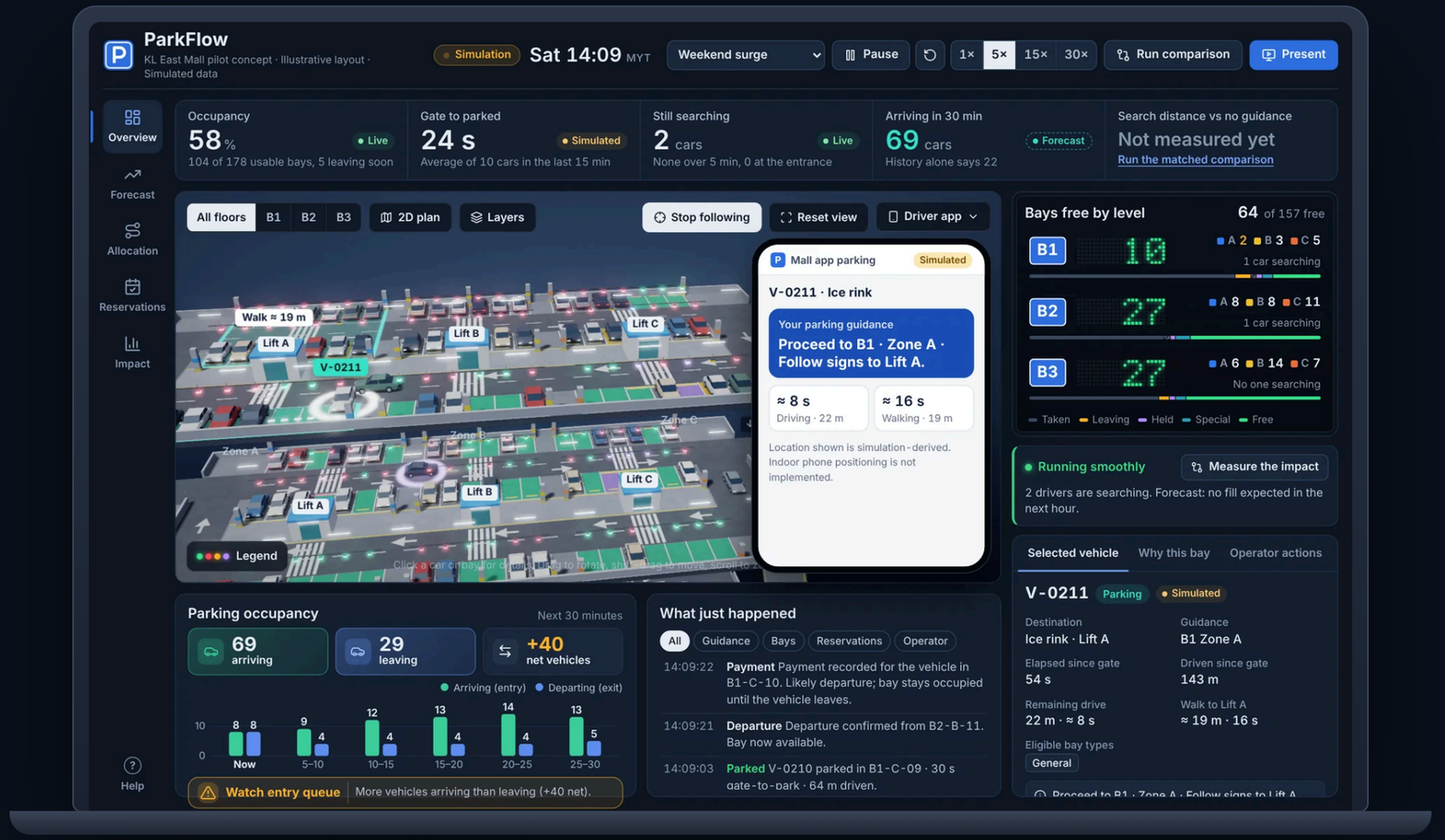
Interactive mobile prototype. Production integration embeds this booking flow directly inside the KL East Mall app.

Operator Dashboard

Track occupancy, gate scans, arriving bookings and exceptions in one view. Use the data to plan staff cover and audit service reliability.

Open the operator screen

Live screens from our working ParkFlow prototype running on simulated weekend traffic.



Priority Zone Access

Normal mode

Walk-ins allowed



Use spare capacity when demand is low.

Demand rises →

← Demand eases

Priority mode

Booking required

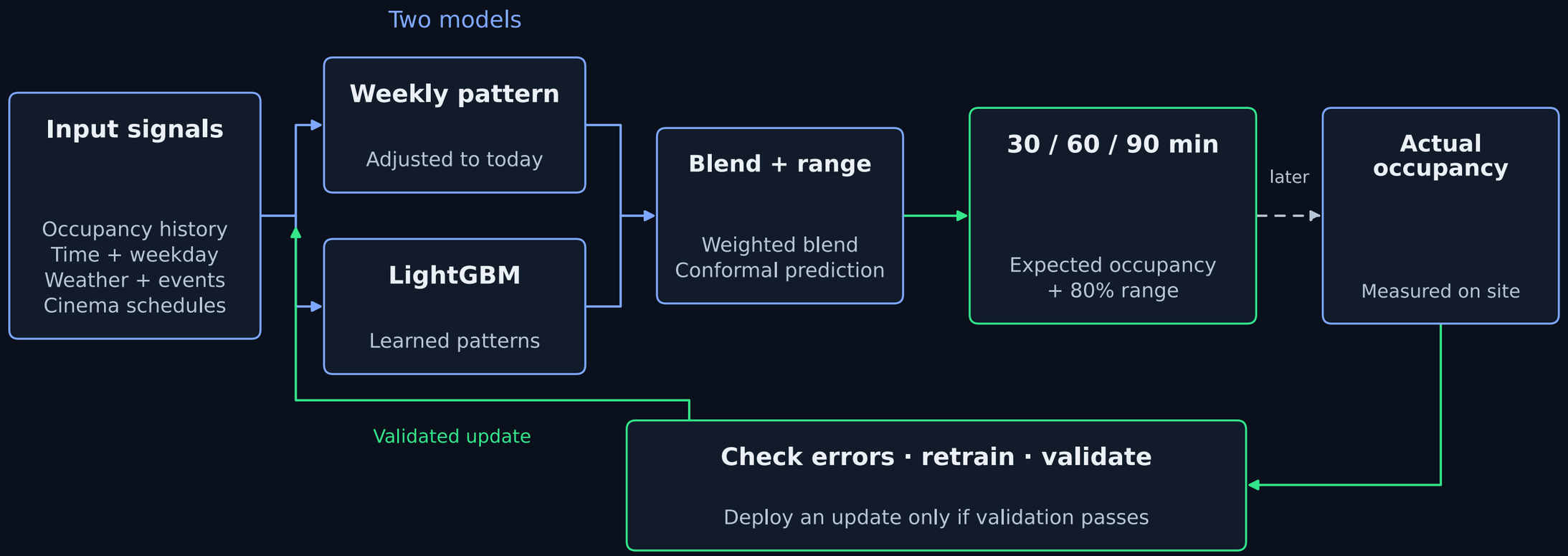


Stop new walk-ins as forecast demand rises.

In both modes: protect arriving bookings + a safety buffer.

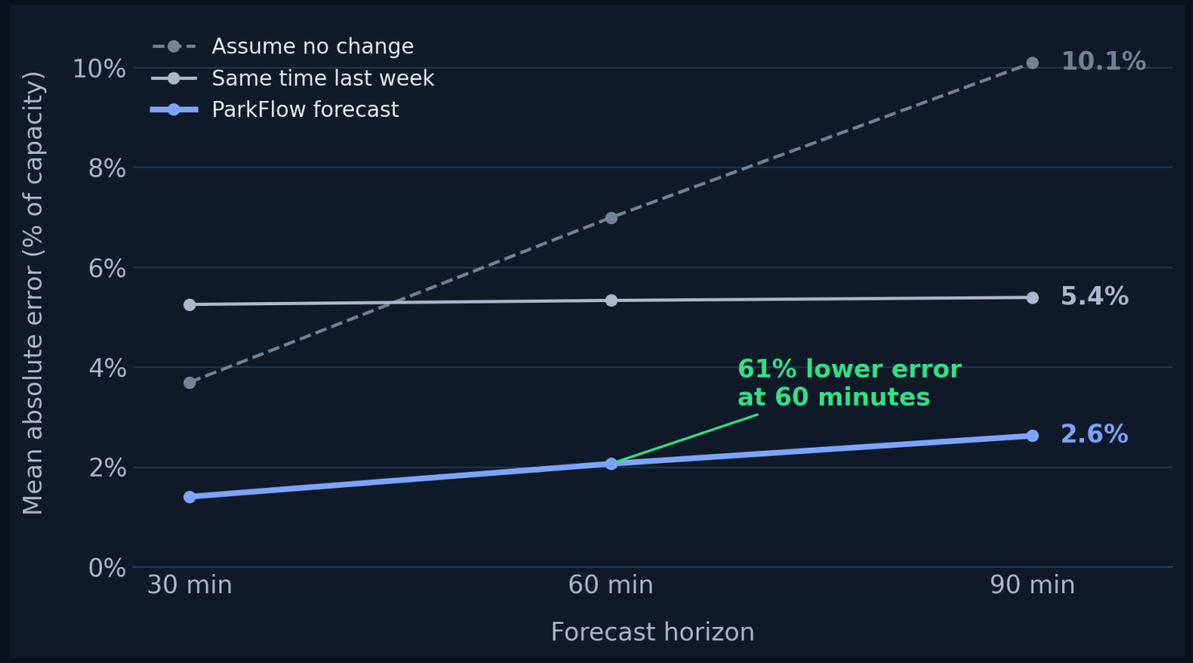
Walk-ins are members without bookings. Verified capacity controls entry. View access rules

How Forecasting Works



Back-test: occupancy history and time features. Weather, events and cinema schedules are proposed pilot inputs.

Forecast Accuracy



Historical occupancy back-test across 48,112 forecasts on 30 real car parks over six weeks.

61%

lower error at 60 minutes

Compared with relying on the same hour last week—reducing average 60-minute error to roughly two bays per 100.

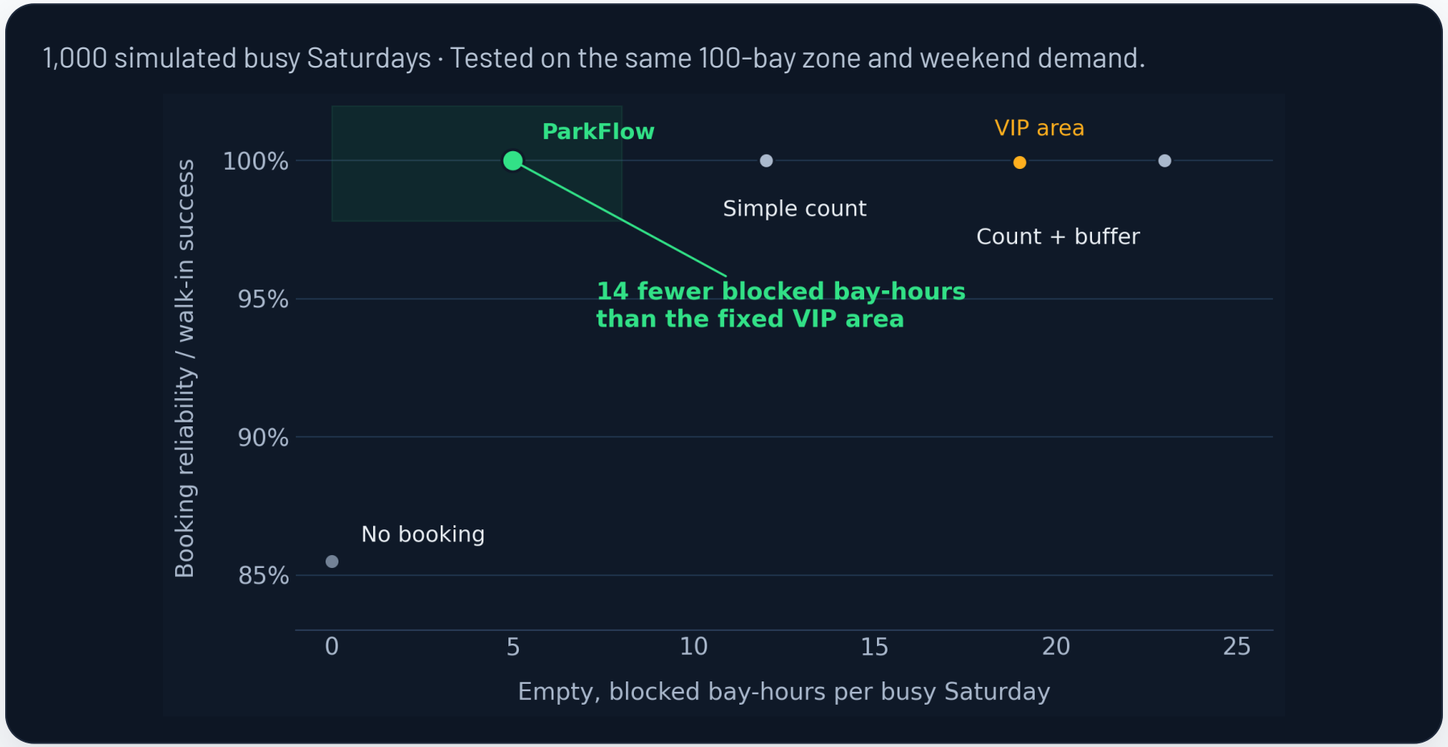
Proven in back-testing

Strong 30-to-90-minute accuracy across 30 real-world car parks.

To validate at KL East

Accuracy by level and zone. Cinema schedules are a proposed additional input.

100% of booked members got a bay—with 74% less idle space.



100%

of booked members got a bay with ParkFlow, compared with 86% without booking control.

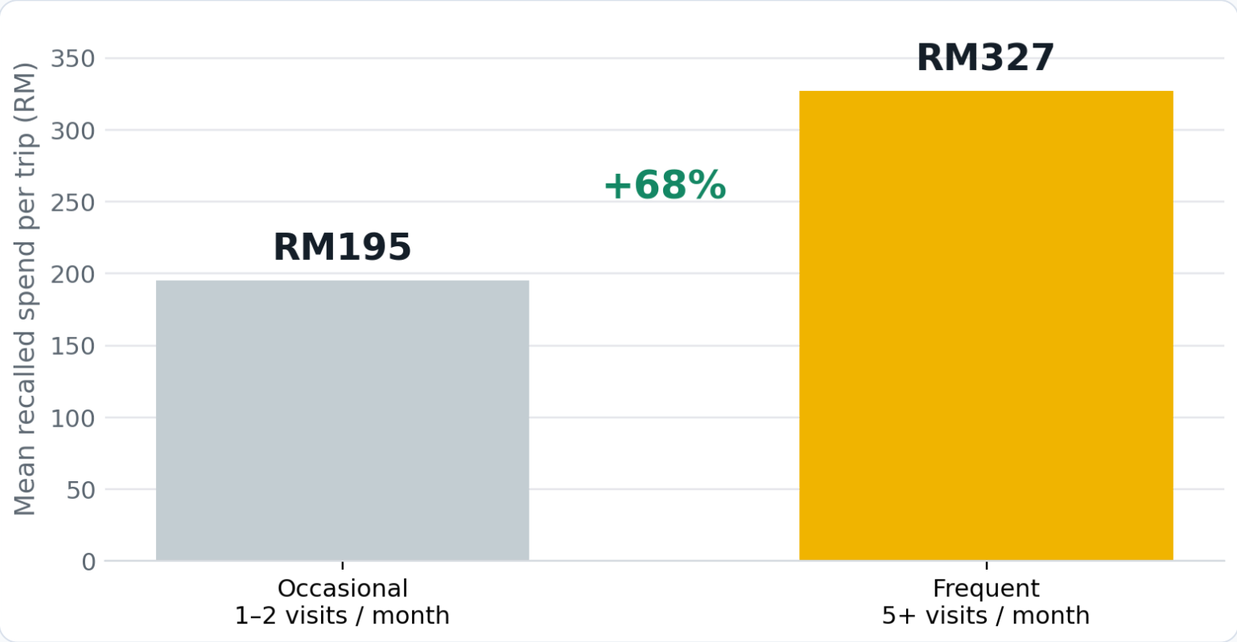
5 vs 19

empty, held bay-hours per Saturday—freeing 14 extra bay-hours compared with a fixed VIP zone.

95% vs 91%

peak zone utilisation between 5:00 pm and 9:00 pm.

Business Impact



Source: Vase.ai Malaysia Shopping Mall Report 2026 (n = 500), average self-reported spend per visit.

Market benchmark

Why frequent visitors matter

Shoppers visiting five or more times a month report RM327 per trip, compared with RM195 for occasional visitors.

Commercial hypothesis

Booking income + repeat visits

Member fees provide direct income. Retained visits create additional mall value only where rental terms capture extra tenant sales.

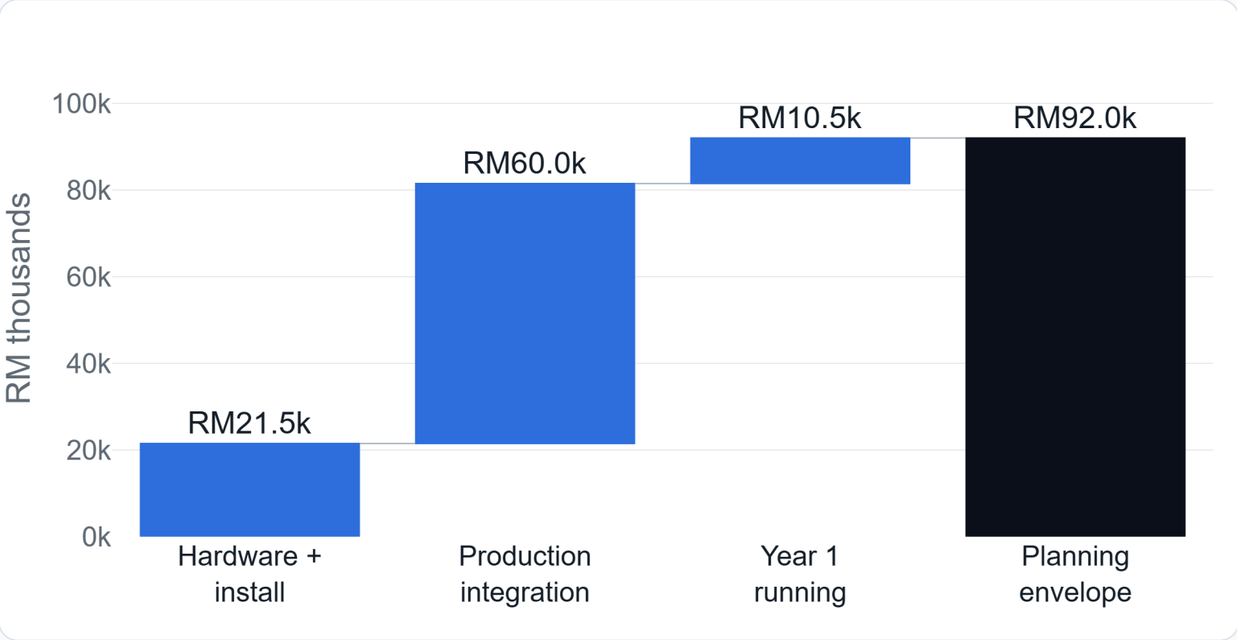
Proposed operating terms

Align the parking operator

A proposed 20% booking-fee share rewards delivery. Confirm responsibilities, settlement and staffing with the operator.

Pilot Investment

RM81.5k upfront plus RM10.5k annual running costs: RM92.0k in first-year funding.



Planning budget based on current Malaysian list prices and engineering estimates prior to final site survey quotations.

RM21.5k

Two smart ANPR cameras, compact parking guidance display, blind-spot cameras, cabling, mounting and commissioning.

Sources: AAC Group Hikvision Price List 2025; Viskou Malaysia DS-TVL224 display. Final installed price requires quotations.

RM60.0k

One-off production build to embed booking, plate verification, and operator controls into the mall app.

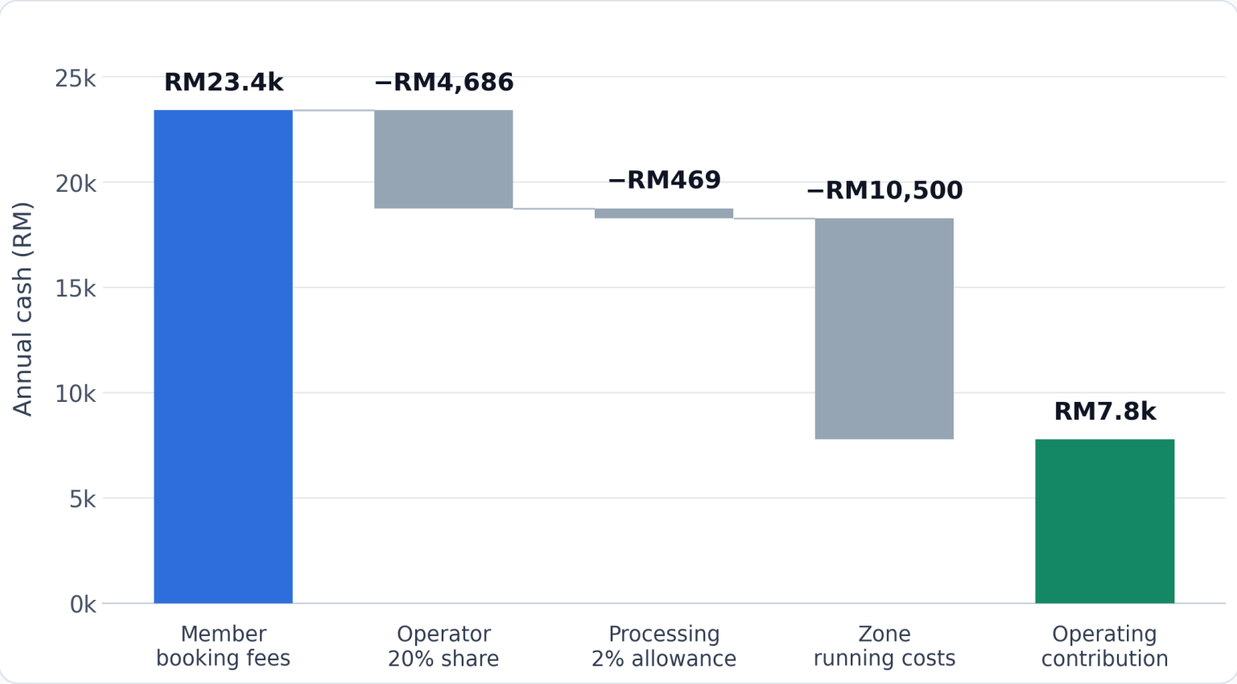
RM10.5k

First-year cloud hosting, WhatsApp/push notifications, and routine maintenance at KL East.

Pre-launch sign-off: Confirm B1 boundary layout, lock in contractor quotes, and agree contingency with SDP procurement before capital commitment.

Booking Revenue & Return

Illustrative annual income from RM4 weekday and RM5 peak bookings.



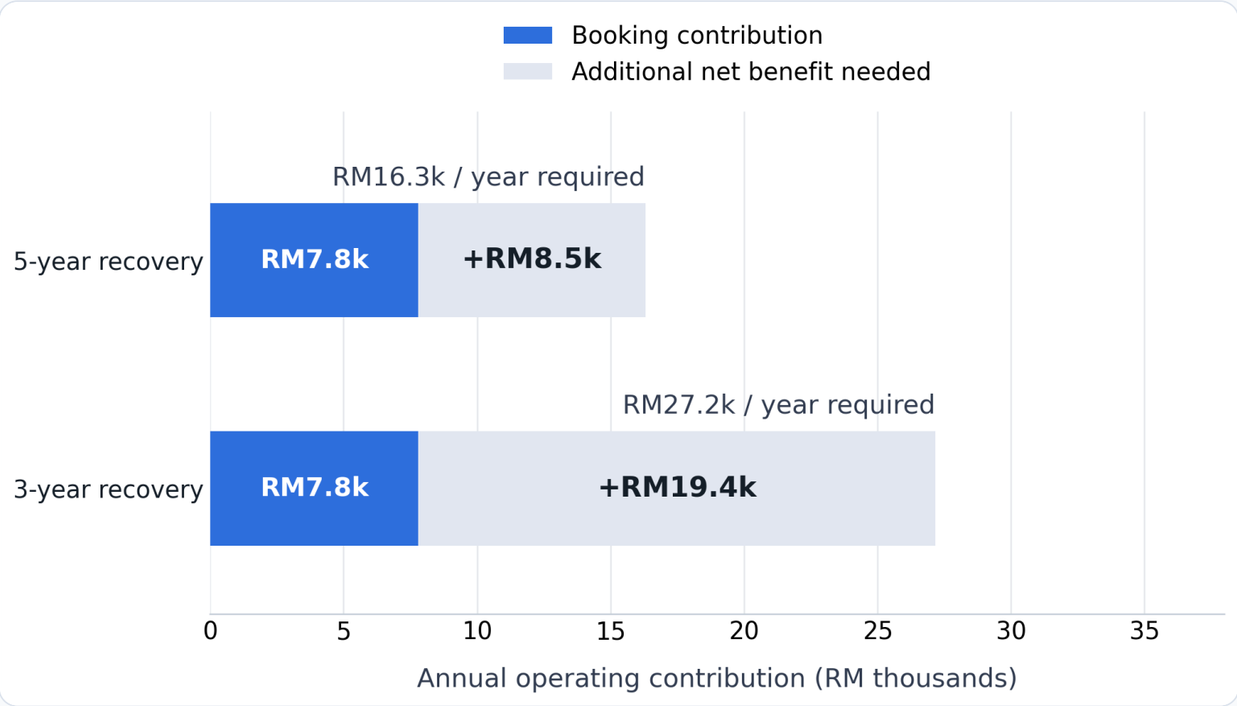
Illustrative demand: 30 paid bookings per peak day × 104 days, plus 7.5 per weekday-priced day × 261 days. Holiday mix to validate.

RM23.4k / year	Gross member booking fees at the proposed RM4 / RM5 prices.
RM18.3k / year	SDP receipts after a proposed 20% operator share and 2% processing allowance.
RM7.8k / year	After RM10.5k zone running costs, before central staffing and tax.
~10.5 years	Simple payback on RM81.5k upfront funding from booking fees alone at this steady annual demand.

Pilot test: confirm paid demand and all delivery costs. Sponsorship and additional turnover rent are separate upside cases.

Investment Conditions

Booking fees alone leave a capital-recovery gap. The pilot must test what can close it.



Illustrative simple recovery of RM81,500 upfront. Three and five years are comparison cases, not SDP's approved hurdle.

Measure paid demand

Validate booking volumes, price acceptance, refunds and settlement costs.

Test additional net value

Retained visits, contracted sponsorship or operating savings count only when evidenced and captured by SDP.

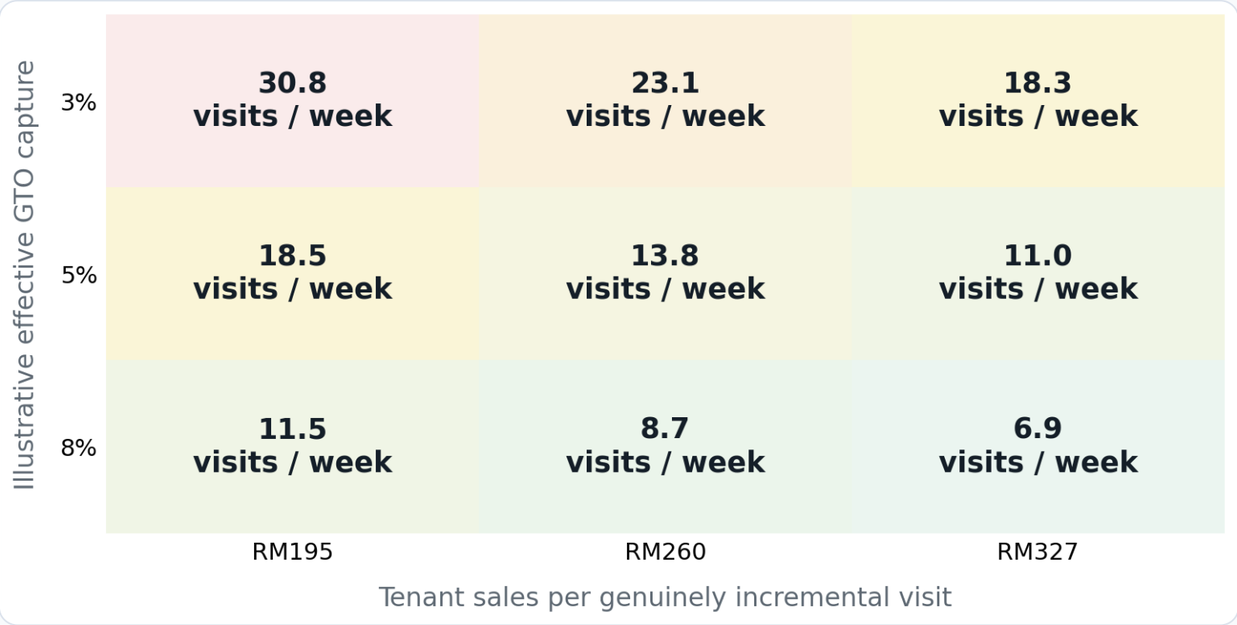
Apply full costs

Add central support, staffing and contingency before making the scale decision.

These are pre-tax, undiscounted illustrations. Additional net benefits must cover their own delivery costs; no sponsorship or retention uplift is assumed here.

Retention Sensitivity

An illustrative RM9,360 annual contribution requires 7–31 additional visits a week, depending on spend and mall capture.



Incremental weekly visits needed across basket sizes and effective GTO capture rates, calibrated with SDP finance during co-design.

Observed **Basket size**

Malaysian mall benchmarks place average trip spend between RM195 and RM327.

Source: Vase.ai, Malaysia Shopping Mall Report 2026, n = 500; self-reported spending.

Assumed **Mall capture**

Modeled across a 3% to 8% effective GTO rent and ancillary capture range.

Measured **Pilot uplift**

Track net additional member visits rather than raw app clicks or reservations.

Decision **Rollout gate**

Scale across the portfolio once measured retention clears SDP's return hurdle.

Payback: validate SDP's rental terms and measure additional visits before estimating capital recovery.

Go-to-Market

Economic buyer

SDP mall / asset management

Approve the pilot budget with finance, using member value and full-cost economics as the investment case.

Pilot champion

KL East GM + Loyalty

Recruit an eligible member cohort through the existing mall app and on-site communications. Test RM4 / RM5 paid demand.

Delivery partners

Operator + Digital / IT

Agree gate integration, data access, service responsibilities and the proposed 20% operator fee share.

First sale A bounded KL East pilot → **Next buyer** Qualified SDP mall teams, using measured pilot results

Proposed approval route, subject to SDP's organisation and procurement process. External licensing remains a later option.

Pilot Plan

Prepare

Survey the zone, obtain quotes, agree the baseline and integrate membership, payments and gate verification.

Run an eight-week live trial

Start with a defined member cohort. Track bookings, entry reliability, paid demand and operating effort.

Review and decide

Compare results with the baseline and full costs. SDP chooses to expand, adjust or stop.



Concept illustration, not a surveyed KL East layout. Final bay count and entry/exit verification depend on the site survey.

Horizon 1: 0–6 months overall. The eight weeks describe the live trial; preparation and review sit around it.

Pilot Success Measures

Agree the baseline and pass criteria before the eight-week live trial.

Member promise

Reliable entry

Track the share of booked arrivals that enter and park without delay, logging every exception.

Member response

Paid demand and repeat visits

Measure willingness to pay, repeat use and any increase in member visits.

Capacity efficiency

Capacity use

Measure peak zone utilisation and how few empty bay-hours are held back to protect bookings.

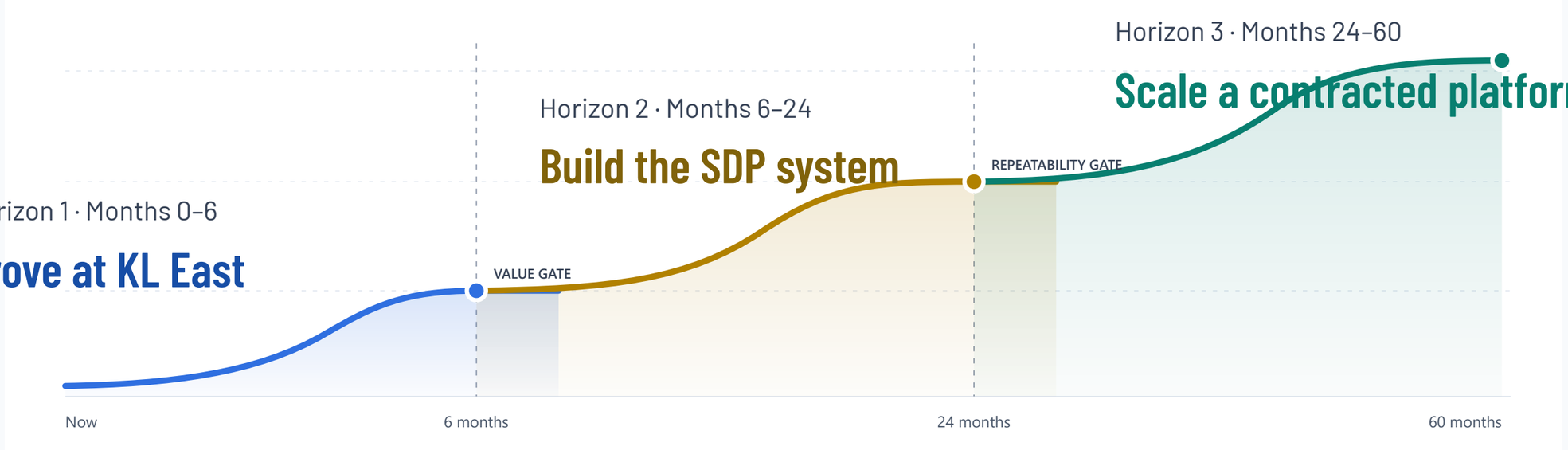
Operational ease

Full delivery cost

Record staffing, support, payment costs and exceptions to test full-cost economics.

Target thresholds are agreed with SDP ahead of Week 1, ensuring an objective scale decision at Week 8.

Three Horizons



Decision pack

Complete site validation, an eight-week live trial, the operating SOP and a go, adapt or stop investment decision.

Portfolio operating system

Qualify and sequence SDP sites, standardise app and operator integration, centralise service levels, and separate Melawati's JV economics.

Commercial platform

Package APIs, data governance, onboarding and support for signed managed-service or licence agreements with mall groups.

Value gate at Month 6; repeatability gate at Month 24. Planning horizons, not revenue forecasts. Sources: SDP FY2025 results briefing; Melawati Mall JV disclosure.

Pilot Decision

Approve a bounded KL East pilot

Confirm the site, operating partner and member cohort. Commit funding only after quotations and the success criteria are agreed.

- Supplier quotes, integration scope and contingency
- Operator agreement and payment settlement
- Service thresholds and full-cost investment hurdle

Planning funding

Upfront build + hardware	**RM81.5k**
Annual zone running cost	**RM10.5k**
First-year envelope	**RM92.0k**

Before contingency and confirmed central support costs. An eight-week live trial sits inside Horizon 1.

Expand only when measured member demand, reliable entry and complete economics clear SDP's agreed thresholds.

Review investment conditions

PARKFLOW · TEAM HEXA

Thank You

Priority parking, promised to members—and proven before it scales.

Questions & discussion

Supporting appendix follows

Value Proposition

01

Member benefit

Eligible members reserve access to pooled Priority Zone capacity right beside the B1 lifts.

02

Automated control

Entry and exit plate cameras protect active bookings and release capacity the moment a car leaves.

03

Auditable proof

SDP tracks utilisation, fulfilment reliability, shopper satisfaction, and repeat-visit uplift in one view.

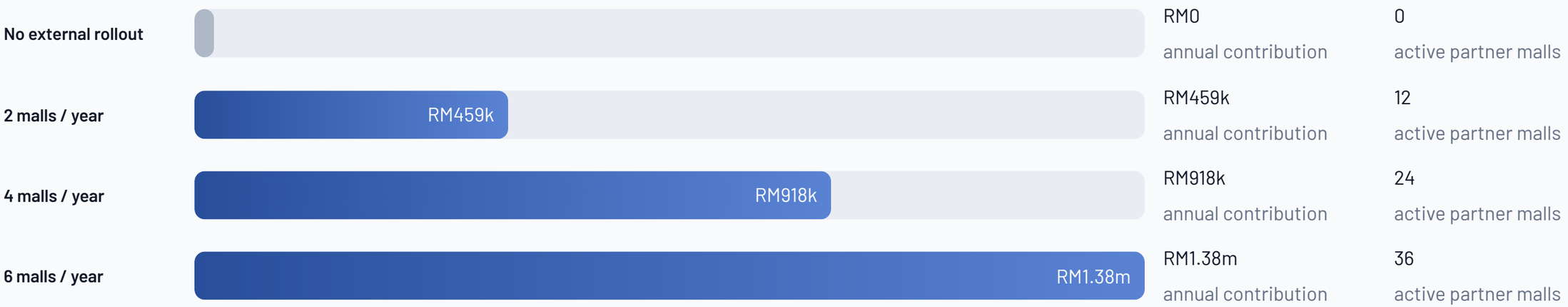
Members	A practical weekend perk they feel on every trip to the mall.
Mall operations	One live, auditable log of arrivals, departures, no-shows, and exceptions.
Loyalty team	A differentiated perk that rewards high-frequency spenders without discounting retail goods.
SDP portfolio	A blueprint ready to replicate across SDP malls once KL East clears its value gate.

Direct income: eligible members pay RM4 on weekdays or RM5 at peaks. Sponsorship, passes and external licensing are separate options.

APPENDIX · COMMERCIAL SENSITIVITY

Partner pace changes the scale case—not the pilot decision.

Illustrative Year 7 annual partner contribution under four post-pilot sign-up rates.



Transparent model: sign-ups begin in Year 2; Year 7 active malls equal six years of additions. Contribution is RM38,250 per mall: RM45,000 target annual fee less RM6,750 cloud/API allowance (1.5 zones × RM4,500).

Decision use: this is a sensitivity, not a forecast. Replace fee, zone count, churn and cloud usage with signed commercial terms before any rollout investment.

Existing CCTV can read free bays accurately enough for a pilot.

99.05%

of spots read correctly, on 4 cameras the AI had never seen before (78,270 photos from a public car park dataset).

99.1%
accurate car count per camera. Our trial target is 95%.

96.1%
of camera views counted within one car

1 ms
to check each spot, even on a normal laptop

- Sunny 99.2%
- Overcast 99.3%
- Rainy 98.6%

To be honest: this was an outdoor car park. Basements are darker, so we will test again on B1 at KL East Mall during the trial.

We tested every option across 1,000 simulated busy Saturdays.

We can't wait for 1,000 real Saturdays, so we built them in a computer and tried every option on exactly the same days.

1. **Create a busy day**
   Crowds at lunch and after 5 pm, most shoppers staying about 1.5 hours, 60 bookings, and some people who never show up.
2. **Try every option on it**
   No booking at all, a fixed VIP area, a simple counting rule, and ParkFlow, on the same 100 spots.
3. **Repeat 1,000 times**
   Then check: did everyone who booked get a spot? How long did spots sit empty while other drivers were turned away?



1,000 busy Saturdays, one dot each

Even when everything goes wrong at once, 99.6% of bookings hold.

Baseline | Stays 20% longer | Higher show-up rate | 10% ignore sign | All three at once

Control method	Bookings honoured	Idle held bay-hours
Fixed VIP zone	99.94%	143 h
Counting rule + 5-bay buffer	100%	127 h
Simple counting rule	100%	62 h
ParkFlow	100%	33 h

99.6%

of booked members still get a bay when longer stays, higher turnouts, and sign violations happen at the same time.

93.4%

booking fulfilment for a fixed VIP zone under the exact same stress test.

A strict counting rule also protects bookings, but locks up nearly four times as many empty bay-hours.

APPENDIX A1 · PILOT SCORECARD

Six clear questions decide whether ParkFlow scales after Week 8.

Dimension	Pre-pilot baseline	Pilot measurement	Pass condition
Member promise	Current weekend parking search experience	Share of booked arrivals that enter and find an open bay	Fulfilment reliability meets or exceeds the agreed SLA
Member demand	Peak member footfall and B1 occupancy	Active users, booking conversion, and repeat reservations	Consistent uptake from target loyalty tiers during peak hours
Loyalty value	Baseline shopper satisfaction and visit frequency	Pre/post member survey and tracked revisit frequency	Measurable lift in repeat visits among participating members
Operations	Current marshal workload and ramp incidents	Manual overrides, gate disputes, and staff minutes per day	Runs smoothly within existing concierge and marshal staffing
Data & privacy	Existing gate and CCTV counter accuracy	Reconciled entry, exit, occupancy, and booking logs	Audit-ready records compliant with the 2024 PDPA amendments
Unit economics	SDP loyalty retention budget and hurdle rate	Verified retention uplift vs. quoted build and run costs	Projected return clears SDP's portfolio investment threshold

Pass thresholds are locked in with SDP management prior to Week 1 so pilot evaluation stays objective.

APPENDIX A2 · PILOT COST BREAKDOWN

Every pilot cost line is transparent and tied to a site-survey check.

Cost item	Planning estimate	Basis of estimate	Pre-approval action
Two smart ANPR cameras (installed)	RM6,000	2 × RM1,918 list price plus RM2,164 mounting and commissioning allowance Source: AAC Group Hikvision Price List 2025, DS-TCG205-E.	Confirm B1 entry/exit sightlines and installer quote
Compact LED parking guidance display	RM4,000	RM2,033 listed display plus RM1,967 mounting, tax and commissioning allowance Source: Viskou Malaysia, Hikvision DS-TVL224-4-5Y.	Confirm viewing distance, mounting, brightness and power feed
Separate edge AI boxes	RM0 base	Plate inference is built into the selected ANPR cameras; use existing NVR or cloud services for the bounded pilot	Add a deployment-grade edge gateway only if SDP IT or privacy review requires one
Blind-spot aisle cameras	RM4,000	Four-camera hardware and mounting allowance benchmarked to current Hikvision dealer pricing Source: AAC Group Hikvision Price List 2025; exact count remains site-dependent.	Reuse existing CCTV coverage where suitable; verify remaining blind spots during survey
Cabling & electrical works	RM7,500	Engineering installation estimate	Obtain contractor quote for PoE runs and conduit
Production software integration	RM60,000	Prototype-to-production engineering scope	Confirm KL East Mall app API, SSO, and DPO sign-off
Year 1 operating & cloud cost	RM10,500	Cloud hosting, messaging, and hardware upkeep	Validate message volume and support SLA with SDP IT
Total pilot planning envelope	**RM92,000**	RM81,500 upfront plus RM10,500 Year 1 operations, before contingency	Replace allowances with binding quotations before approval

APPENDIX A3 · EVIDENCE LEDGER

What is already proven vs. what the KL East pilot will validate in the field.

Area	Current status	Evidence in hand	Validated during the KL East pilot
Higher spend by frequent shoppers	Observed	Vase.ai Malaysia Shopping Mall Report 2026 (n = 500)	Actual spend and revisit uplift among KL East members
Booking reliability under stress	Simulated	1,000 simulated busy Saturdays across 5 stress scenarios	Live B1 gate fulfilment and driver compliance rates
60-minute occupancy forecasting	Back-tested	48,112 forecasts tested across 30 real-world car parks	Calibration to KL East cinema and dining turnover peaks
RM92.0k pilot budget	Costed	Current Malaysian component prices and engineering build scope	Binding local contractor and system-integrator quotes
Member retention economics	Pilot test	Weekly visit sensitivity matrix across GTO capture rates	Matched pre/post member visit frequency with SDP Loyalty
Portfolio & platform rollout	Stage-gated	Three-horizon expansion roadmap	Repeatability across Melawati, Elmina, and KLGCC malls

Published market data · Simulated, back-tested, or costed · Validated live in the KL East pilot

Priority Zone Access Rules

Member eligibility	Both modes remain members-only. A walk-in is an eligible member without a booking.
Admission control	Before a walk-in enters, verified empty capacity must cover outstanding arriving bookings, the safety buffer and that new admission.
Mode switching	Forecast demand guides the switch to Priority mode or a return to Normal mode. A predicted departure cannot create bookable capacity.
Existing parked members	Priority mode restricts new walk-in entry. Members already inside stay until they leave.
Pilot validation	Agree buffer size, switching thresholds and operating responsibilities before launch. Local forecasts and the switching policy require field validation.